SAECOIN

BACKUP & RECOVERY POLICY

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1. Purpose

This document defines backup, data protection, and recovery requirements for SAECOIN software and infrastructure.

The primary objectives are:

- Protect wallet data.
- Prevent permanent loss of private keys.
- Preserve blockchain and node data.
- Protect project source code.
- Enable recovery after hardware failure.
- Reduce the risk of accidental deletion.
- Maintain operational continuity.

2. Scope

This policy covers:

1. Wallet private keys.
2. Wallet seed phrases.
3. Encrypted wallet files.
4. Wallet configuration.
5. Full-node blockchain databases.
6. Node configuration files.
7. Mining configuration.
8. Explorer databases.
9. Server configuration.
10. Source code and documentation.
11. Release signing infrastructure.
12. Testnet and operational records.

3. Backup Principles

All SAECOIN backups should follow these principles:

3.1 Confidentiality

Sensitive wallet information must remain private and encrypted where appropriate.

3.2 Integrity

Backups must be protected against corruption and unauthorized modification.

3.3 Availability

At least one recoverable backup should remain accessible if the primary device fails.

3.4 Separation

Backups should not all depend on the same physical device or location.

3.5 Verification

A backup is not considered reliable until its recovery process has been tested.

4. Wallet Backup Policy

Wallet backups are among the most important parts of SAECOIN.

A wallet may contain private keys that authorize spending.

Loss of the required private keys can result in permanent loss of access to associated funds.

4.1 Seed Phrase

If the final wallet design uses a seed phrase, the wallet must provide a secure backup procedure.

Requirements:

- Display the seed only during a trusted wallet setup or recovery process.
- Warn users never to share it.
- Encourage offline backup.
- Require confirmation that the backup has been recorded correctly.
- Never send the seed phrase to a remote server.
- Never include seed phrases in application logs.
- Never request seed phrases through support messages.

The seed format and word-list standard remain undecided.

4.2 Private Keys

Private keys must never be stored in plaintext unless a specifically reviewed design justifies a temporary in-memory representation.

Requirements:

- Use cryptographically secure key generation.
- Protect stored keys using an approved encryption design.
- Avoid unnecessary key exposure in memory.
- Never upload private keys to analytics or remote services.
- Never commit private keys to Git repositories.

4.3 Encrypted Wallet Files

If wallet files are used, the application should support encrypted backups.

The final encryption algorithm, key derivation function, parameters, and file format must be selected through security review.

Users should be informed that losing both the wallet file and its recovery credentials may make recovery impossible.

5. Wallet Recovery

Wallet recovery must be tested before any production wallet release.

Expected recovery process:

1. Install an authentic SAECOIN wallet.
2. Select the recovery option.
3. Enter the approved recovery information.
4. Validate the recovery data locally.
5. Reconstruct the wallet keys.
6. Restore wallet addresses.
7. Synchronize blockchain information.
8. Verify that expected funds and transactions are visible.

Recovery must not depend on a centralized SAECOIN server holding users' private keys.

The exact recovery procedure depends on the final wallet standard.

6. Full Node Backup

Full nodes maintain blockchain data and operational configuration.

Node operators should distinguish between:

- Blockchain data.
- Wallet data, if the node manages a wallet.

- Configuration files.
- Peer information.
- Logs.
- Application binaries.

Blockchain data may be downloadable again from the network, depending on network availability and integrity.

Private wallet data must never be treated as replaceable blockchain data.

6.1 Node Backup Requirements

The node software should document:

- Data directory location.
- Configuration file location.
- Database format.
- Backup procedure.
- Restore procedure.
- Database compatibility requirements.

6.2 Database Backup

Database backups must use a method compatible with the selected database engine.

If RocksDB is selected, backup procedures must account for database consistency and file-locking behavior.

Copying a live database directory without a safe snapshot or shutdown procedure must not be assumed to produce a valid backup.

RocksDB remains a provisional technology choice.

7. Mining Configuration Backup

Mining configuration may include:

- Mining software settings.
- Node connection details.
- Performance configuration.
- Device-specific settings.

Private wallet keys or sensitive authentication credentials must not be embedded in public mining configuration files.

Mining configuration should be recoverable without exposing wallet secrets.

8. Explorer and Server Backup

If SAECOIN operates explorer or API infrastructure, operators should back up:

- Database contents.
- Application configuration.
- Deployment configuration.
- TLS and service configuration.
- Operational documentation.
- Relevant monitoring configuration.

Private server credentials must be stored separately using secure secret-management practices.

Explorer data should be rebuildable from verified blockchain data where practical.

9. Source Code and Documentation Backup

Project source code and documentation should be stored in version-controlled repositories.

Recommended protections:

- Private backup copy of the repository.
- Regular remote repository synchronization.
- Protected branches.
- Access controls.
- Multi-factor authentication.
- Release tags.
- Periodic export of important project documentation.

GitHub should not be treated as the only backup location.

The project should maintain an independent backup of important source code and documents.

10. Backup Frequency

The following is a proposed operational schedule.

Data Type| Suggested Backup Frequency
Wallet recovery information| At creation and after approved changes
Encrypted wallet backup| After wallet changes
Source code| After meaningful development changes
Project documentation| After every approved revision
Node configuration| After every configuration change
Explorer database| Daily, if operational
Server configuration| After every meaningful change
Operational records| Daily or after significant incidents

Backup frequency must be adjusted to operational risk and data-change frequency.

11. Backup Storage Strategy

A 3-2-1 backup strategy is recommended where practical:

- Maintain three copies of important data.
- Use at least two different storage types.
- Keep at least one copy in a separate physical location.

For wallet recovery information, physical offline storage may be appropriate.

For sensitive digital backups:

- Use strong encryption.
- Restrict access.
- Protect encryption credentials separately.
- Avoid untrusted cloud storage.
- Avoid sharing backups through ordinary messaging applications.

12. Recovery Testing

Backups must be tested periodically.

Recommended tests:

- Restore an encrypted wallet backup.
- Recover a wallet using its approved recovery method.
- Restore node configuration.
- Restore a database snapshot.
- Rebuild a development environment.
- Restore project source code from an independent backup.

A successful backup operation does not prove that recovery will work.

13. Recovery After Device Failure

If a computer or server fails:

1. Identify affected data.
2. Determine whether wallet keys were stored on the device.
3. Secure available backup copies.
4. Reinstall a trusted operating system and required software.
5. Restore configuration and data.
6. Verify integrity.
7. Resume services only after validation.

If wallet secrets may have been exposed, the incident response process must be followed.

14. Lost or Compromised Wallet

If a wallet device is lost or compromised:

- Determine whether private keys may have been exposed.
- Stop using potentially compromised software.
- Recover the wallet using a trusted device.
- If possible, move funds to a newly generated secure wallet.
- Protect the new recovery information.
- Document the incident.

A recovery phrase cannot reverse an already completed unauthorized transaction.

15. Backup Encryption and Access Control

Sensitive backups must use an approved encryption mechanism.

The project must define and review:

- Encryption algorithm.
- Key derivation method.
- Password requirements.
- Authentication requirements.
- Backup file format.
- Recovery compatibility.

Backup credentials should not be stored together with the encrypted backup unless the design has been reviewed for that risk.

16. Backup Retention

The project should define retention periods for:

- Daily operational backups.
- Weekly backups.
- Monthly archival backups.
- Release records.
- Security incident records.

Retention periods must balance recovery needs, storage cost, and privacy requirements.

Wallet backup retention remains under the user's control.

17. Backup Failure Response

If a backup fails:

1. Record the failure.
2. Identify the affected data.
3. Retry using a verified procedure.
4. Check the integrity of existing copies.
5. Escalate critical failures.
6. Do not delete the previous valid backup until the new backup is verified.

18. Privacy Requirements

Wallet backups may contain highly sensitive financial access information.

The project must not:

- Collect seed phrases.
- Request private keys through support.
- Log sensitive recovery data.
- Expose wallet backups publicly.
- Transmit unencrypted private-key material.
- Include user recovery information in diagnostic reports.

19. Current Project Limitations

At the time of this document:

- The final seed phrase standard has not been selected.
- The wallet backup file format has not been finalized.
- The wallet encryption standard has not been finalized.
- The final database engine has not been selected.
- Production backup infrastructure does not yet exist.
- Recovery testing has not yet been performed.

These items must be finalized before production wallet deployment.

20. Conclusion

This policy establishes the backup and recovery principles for SAECOIN.

Wallet recovery, private-key protection, and data integrity must be treated as critical requirements.

No wallet or infrastructure backup system should be considered production-ready until its recovery procedures have been successfully tested.

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